Case Studies in Thermal Engineering 72 (2025) 106343

Contents lists available at ScienceDirect

Case Studies in Thermal Engineering

journal homepage: www.elsevier.com/locate/csite

Effect of catalyst particle shape on heat and mass transfer in steam methane reforming

Igor Karpilov, Dmitry Pashchenko *

Faculty of Mechanical Engineering, Technion - Israel Institute of Technology, Technion City, Haifa 3200003, Israel
Department of Mechanical Engineering and Guangdong Provincial Key Laboratory of MATEC, Guangdong Technion - Israel Institute of Technology, 241 Daxue Road, Guangdong, Shantou, 515063, China

ARTICLE INFO

Keywords:
Steam methane reforming
Hydrogen production
Heat transfer
Porous media
CFD
Catalysis

ABSTRACT

Steam methane reforming (SMR) is a complex process with several intertwined dependencies. An important factor is heat transfer between the reaction mixture and the catalyst, which greatly affects CH_4 conversion. The present manuscript investigates the influence of catalyst particle shape on heat transfer characteristics during SMR reactions. A numerical model based on the CFD approach was developed to compare different catalyst shapes and determine the optimal one. Five catalyst shapes: sphere, cylinder, Raschig ring, 7-hole cylinder, and trilobe were analyzed under varying operating conditions (inlet velocities of 0.25–1.5 m/s, temperatures of 700–1100 K, and particle diameters of 2.5–20 mm). The results demonstrate that Raschig rings and 7-hole cylinders exhibit the highest total heat flow, consuming up to 276 W under optimal conditions (T = 1100 K, v = 1.5 m/s), while spherical particles with a diameter of 20 mm achieve the highest heat flux (19320 W/m^2) due to their streamlined geometry. In contrast, Raschig rings provide the highest mass-specific heat flow (3520 W/kg), outperforming 7-hole cylinders by 12.5% and trilobe particles by 15%. The study reveals that catalyst performance is governed by three key factors: high surface area, low flow resistance, and streamlined shape. These findings provide critical insights for optimizing catalyst design in industrial SMR applications, balancing reactor volume efficiency (favoring spheres) and economic feasibility (favoring Raschig rings).

1. Introduction

Global demand for hydrogen in 2022 has increased by 3% compared to 2021 and is expected to grow by 1.5 times by the end of this decade [1]. The primary industries that require hydrogen are ammonia and methanol production, refineries, steel, and the iron industry [2]. In addition, hydrogen will be used in new technological areas such as transportation, building, and electricity production. The primary hydrogen production technology is steam methane reforming (SMR) [3,4]. SMR is used to produce 95% of world hydrogen supply [5,6]. This technology is not currently environmentally friendly, as natural gas combustion is used to supply heat for SMR reactions [7]. Also one of the main reaction products is CO_2. Nevertheless, SMR has the potential for sustainable hydrogen production. The heat supply could be changed to an environmentally friendly source such as solar radiation or green power [8,9]. Carbon capture and storage technology could be adapted to sequester one of the main reaction products - CO_2 [10,11].

Steam methane reforming is a complex process that involves several endothermic reactions occurring in the vicinity of the catalyst [12,13]. In order to sustain these reactions, an external heat source is required. The primary fuel components used are

* Corresponding author at: Faculty of Mechanical Engineering, Technion - Israel Institute of Technology, Technion City, Haifa 3200003, Israel.
E-mail addresses: dmitry.pa@technion.ac.il, dmitry.pashchenko@gtiit.edu.cn (D. Pashchenko).

https://doi.org/10.1016/j.csite.2025.106343
Received 11 February 2025; Received in revised form 13 May 2025; Accepted 15 May 2025
Available online 2 June 2025

methane and steam, and the main reaction products are hydrogen, carbon monoxide, and carbon dioxide. Porous catalysts with metal coating are usually used to enhance the reaction kinetics [14,15]. A number of catalysts form a packed bed with a variety of possible topologies [16]. Thus, in the SMR reactor, several simultaneous processes occur: heat and mass transfer between reacting components, reactor walls and catalysts; complex fluid flow through a packed bed; consumption of reactants and reaction products forming; intraparticle diffusion of the flow. These processes are studied separately or concurrently by researchers [17,18].

Each process requires extended research as there are several intertwined dependencies. Pressure drop through the packed bed depends on the particle diameter, packed bed porosity, fluid velocity and density [19,20]. Reaction rates, in turn, depend on flow temperature, pressure, fluid velocity, available surface area, H_2O/CH_4 ratio and catalyst coating type [21,22]. Intraparticle diffusion changes with particle diameter and porosity and significantly decreases the overall H_2 yield due to the mass transfer limitations [23]. The relationships mentioned above show that the SMR process is rather complicated and research of one of its components leads to a better understanding of the whole process.

There is significant scientific interest in SMR reactions, highlighting the importance of research in this field [24,25]. One of the most difficult challenges is coupling the heat and mass transfer inside individual particles with industrial-scale packed bed reaction models [26]. Heat and mass transfer inside the packed bed particles govern the temperature distribution and species molar fraction concentration. Both of these parameters influence the H_2 yield significantly. While the external source heats the reaction mixture, it is cooled by the endothermic reactions [27,28]. The exact influence of the enthalpy of reaction on the heat and mass transfer properties of the flow is of great interest. Therefore, heat and mass transfer inside the SMR reactor is studied extensively [29,30].

Ma et al. [31,32] conducted a study on heat and mass transfer in a high-temperature heat pipe reactor powered by solar radiation. The heat was transferred to the reaction zone mainly by thermal conductivity and a nearly uniform reactor temperature was achieved. Despite achieving a high efficiency of 41%, the SMR reactor experienced heat loss of over 50% due to radiation and convection from its outer surface.

Heat convection is the primary heat transfer mechanism from the heated reactor surface to the reaction mixture. Wang et al. numerically compared [33] the randomly filled packed bed with a multi-channel metal framework structure. Heat transfer analysis was conducted for the mixture inlet and wall temperature at 673 K and 824 K, respectively, under various inlet velocities. The high thermal conductivity of the metal allowed for effective heat distribution in the radial direction. As a result, the novel structure had a heat transfer coefficient that was 2–5 times higher than that of unstructured packing.

A packed bed typically contains catalyst particles of different shapes and sizes [34]. Wu et al. [35] studied the impact of particle diameter on reaction kinetics. The authors measured the heat transfer coefficient and hydrogen yield on structured packed beds. It was found that the total heat transfer coefficient per unit of catalyst surface increased with an increase in particle diameter. However, hydrogen yield increased when the particle diameter was smaller. It is important to note that in this study, the surface area was not kept constant. Therefore, the higher CH_4 conversion observed was due to the increased catalyst surface area. Karthik and Buwa [36] investigated the trade-off between heat transfer and pressure drop. The authors compared packed bed filled with cylinders, trilobes, daisies, hollow cylinders, cylcut and 7-hole cylinders. The results indicated that the best trade-off is observed for the trilobe particles. The externally-shaped particles provide high heat sink and low pressure drop due to the less tortuous path of the flow. The study was carried out for various inlet velocities, whereas the other parameters remained constant. Xu et al. [37] studied the flow dynamics and heat transfer for an exhaust reformer. The reformer performance was evaluated for six catalyst shapes: spheres, one-hole sphere, cylinder, Raschig ring, trilobe and 4-hole cylinder. The one-hole spheres were found to have the highest H_2 production to pressure drop ratio. In addition, it was shown that the hot spot position was moving closer to the inlet with a higher temperature.

Various heating methods also influence the heat distribution inside the packed bed. Lu and Nikrityuk [38] investigated the heat transfer driven by Joule heating in an SMR reactor. The increase in input electric power led to increased temperature and CH_4 conversion. Higher mass flow led to decreased residence time and thus lowered the H_2 yield. The study also found that a linear increase in electric current and mixture mass flow caused a decrease in CH_4 conversion, indicating that mass flow has a greater effect on the reaction rate. Another study by Siavashi et al. [39] utilized reaction mixture pre-heating to improve heat distribution in solar-driven steam methane reforming. The authors re-routed the waste energy of the reaction zone to the pre-heating section. This reactor improvement allowed for a better use of the solar heat and increased the H_2 yield by 20%.

Researchers use three primary methods to study the steam methane reforming process: experimental, analytical and numerical. Extensive experimental studies were conducted to investigate the characteristics of the SMR process [40,41]. Peloquin et al. [25] experimentally investigated the efficiency of an electrically heated 3D-printed microreactor for SMR reactions. Shigarov et al. [42] studied steam reforming of methane-propane mixture for various operating conditions on an experimental setup. An experimental study on a high-temperature tubular reactor driven by solar and electrical heating was conducted by Ma et al. [43]. The authors managed to increase the reactor efficiency up to 45%.

The analytical approach, on the other hand, was used by Pashchenko and Eremin [44] to calculate the temperature field inside a spherical particle for SMR reactions. Pourali et al. [45] implemented an analytical model to study the design and operating parameters of the SMR microreactor. The analytical approach only applies to cases with uniform catalyst particle heating, does not calculate the parameters outside the catalyst particle and only applies to cases with relatively simple geometry [46,47].

Numerical modeling, especially CFD, allows insight into the distribution of all the parameters inside the SMR reactor and is a primary tool for SMR investigation [48,49]. Chen and Yu [50] numerically investigated optimal operating parameters for a microreactor with methanol steam reforming. Amini et al. [20] used CFD to enhance the hydrogen production of a reformer tube. Wu et al. [51] compared a novel packed bed design with an existing one by means of CFD modeling. Karpilov and Pashchenko [52] performed a CFD modeling to study the heat and mass transfer of SMR reactions on a pre-heated packed bed.

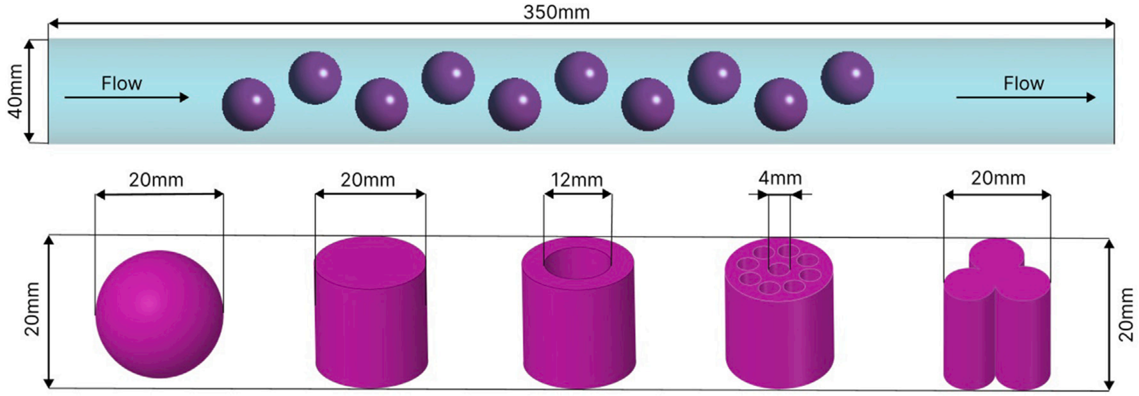

Fig. 1. Computational geometry.

The available research suggests that the heat and mass transfer processes play a crucial role in SMR reactions, but there has been limited study in this area. It is evident that particle shape significantly affects the heat and mass transfer between the reacting flow and catalyst particle. Still, there is a lack of detailed understanding of the effect of particle shape, as well as reactor operating conditions and design parameters on the heat and mass transfer mechanism. This article aims to describe the influence of catalyst particle shape on heat transfer characteristics during SMR reactions. To quantify this effect, the SMR reactor operation under various residence time, temperatures, reaction mixture composition and particle diameter was analyzed via the CFD modeling. Five catalyst particle shapes were compared under various operating conditions. As a result, the guidelines were formulated for the choice of catalyst particle from the heat and mass transfer perspective.

2. Methodology

2.1. Geometry and mesh

In the present investigation, the heat and mass transfer process was studied in the course of steam methane reforming reactions. The geometry used for the numerical simulation is shown in Fig. 1. The geometry consists of a cylindrical tube with diameter $D = 40$ mm and length $L = 350$ mm filled with ten catalyst particles of the same size but different shapes. That number of catalysts is chosen to enhance the SMR reactions and obtain more precise differences between the results. The shape of catalyst particles was chosen according to the most widely available commercial Ni-Alumina catalysts [53]. Space between each catalyst is left to enable the free flow between the particles. That condition allows to better study the heat and mass transfer process on the catalyst of various shapes [54]. The significant distance of the particles from the wall is created to lower the influence of the boundary layer on the flow around the particles. Space before and after the catalysts was created for the flow development and vortex dissipation, respectively.

The meshing procedure was carried out for polyhedral elements as they allow a decrease in the number of elements and computational time compared to tetrahedral elements [55]. The mesh structure is presented in Fig. 2. It comprises four zones: freeflow zone, tube boundary layer, transition zone and internal catalyst zone. The inflation procedure created the boundary layer on the tube wall. The Freeflow zone has the largest elements and is located in tube space away from the catalyst. The transition zone contains the smallest mesh elements to better capture the mass, momentum, temperature and mixture species gradient. The catalyst interior is filled with elements of average size as gradients are less steeper here than in the transition zone.

The mesh independence study was carried out both for polyhedral and tetrahedral elements and is shown in Fig. 3. The sensitivity of CH_4 average molar fraction (Fig. 3a) and the average temperature on the outlet (Fig. 3b) to the number of mesh elements was tested. The test was performed for cylindrical particles with $d_p = 20$ mm under the inlet velocity of 1.5 m/s and temperature of 1100 K. The mesh size varied for polyhedral elements from 848 425 to 5 556 720 elements and for tetrahedral elements from 1 264 802 to 12 550 778 elements. The number of tetrahedral elements was about four times higher to achieve the same solution accuracy. The deviation between mesh with 3 872 600 and 5 556 720 polyhedral elements was less than 0.3%. Therefore, 3 872 600 elements were chosen for the present investigation. The computational time on tetrahedral mesh took 23% longer. As a result, the polyhedral mesh was chosen.

2.2. Governing equations

The proposed CFD model contains a flow zone with a porous catalyst. To model the mixture flow, heat transfer and chemical reactions the following equations were solved: momentum, continuity, energy and species transport. The solution method is finite volume, and flow is considered 3D for a steady-state simulation. The main assumptions of this study are the following:

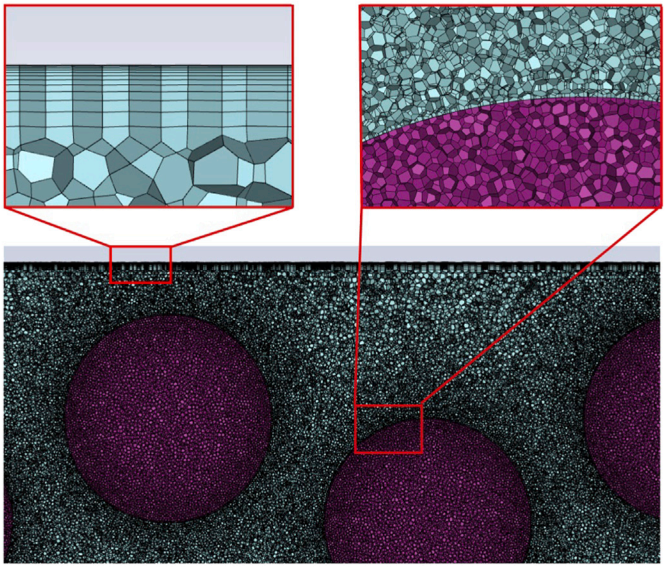

Fig. 2. The computational mesh and its structure.

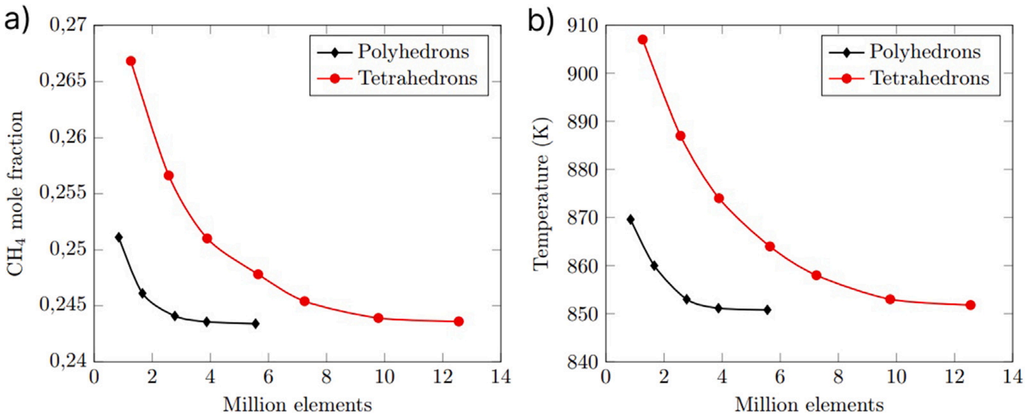

Fig. 3. Mesh independence study for cylindrical particles $d = 25$ mm (a) CH_4 molar fraction on outlet; (b) Average temperature on outlet.

- Steady-state formulation.
- Incompressible ideal gas flow.
- Adiabatic reactor walls condition.
- The catalyst is treated as a homogeneous porous media.
- The radiation heat transfer was neglected.
- Laminar flow inside the catalyst particles.
- No carbon deposition on the catalyst surface.
- Fick's law governs species diffusion, with Soret thermal diffusion included but turbulent diffusion neglected inside particles.
- No particle-to-particle interactions. Heat and mass transfer between adjacent particles is simplified by spacing them apart in the model.

The radiation was neglected in this study as it becomes significant at very high temperatures (above 1200 K). The studied range (700–1100 K) is below the threshold where radiative heat flux has a significant influence on heat transfer. Other CFD studies of SMR or packed-bed reactors in this temperature range neglect radiation without significant loss of accuracy [36,51].

The proposed CFD model solves separate sets of governing equations for the free-flow regions and porous catalyst zones. Below, the equations for the region outside catalyst particle are defined. The continuity equation for a steady 3D flow is written in the form:

$$\nabla \cdot (\rho \vec{v}_{av}) = 0 \tag{1}$$

where ρ and $\vec{v}_{av}$ are mass average density and velocity of the mixture.

The momentum conservation equation is used in the form:

$$\nabla \cdot (\rho\, \vec{v}_{av}\, \vec{v}_{av}) = \nabla \cdot (\bar{\bar{\tau}}) - \nabla \cdot p + \rho\vec{g} + S_i, \tag{2}$$

where p is mixture pressure, g is gravitational acceleration value and S_i is a source term that contains the external body force, which in the present model is inertial and viscous resistance caused by flow through porous media.

The stress tensor $\bar{\bar{\tau}}$ used in Eq. (2) is given as:

$$\bar{\bar{\tau}} = \mu\vec{v}_{av}^{\mathrm{T}} + \mu\nabla\vec{v}_{av} - \frac{2}{3}\mu\nabla \cdot \vec{v}_{av} I, \tag{3}$$

where μ represents mixture viscosity, I is a unit tensor, which is a part of the volume dilatation term.

The source term S_i in the momentum equation (Eq. (2)) contains the Darcy term and inertial loss term. For the homogeneous porous catalyst the S_i takes the following form [56]:

$$S_i = -v_i \cdot \left(\frac{\mu}{\alpha} + C_2 \frac{1}{2}\rho|v_{av}|\right), \tag{4}$$

where α and C_2 are the permeability and inertial resistance factor, respectively. It should be noted that S_i equals 0 for the zones outside of the catalyst.

The inertial resistance factor α and permeability C_2 were determined using the widespread semi-empirical Ergun equation, which is applicable for cases with a wide range of Reynolds number and packed bed configurations [20,57]. The Ergun Equation is written in the following form:

$$\frac{\Delta p}{L} = \frac{150\mu}{d_p^2}\frac{(1-\epsilon)^2}{\epsilon^3}v_s + \frac{1.75\rho}{d_p}\frac{(1-\epsilon)}{\epsilon^3}v_s^2, \tag{5}$$

where v_s is a superficial velocity and d_p is the mean particle diameter and ϵ is media porosity.

The Ergun equation consists of two terms on the right-hand side. The first term accounts for viscous resistance mostly encountered in laminar flow and is called the Blake–Kozeny equation. The second term describes the kinetic energy loss that is present in turbulent flows and is called the Burke–Plummer equation. The coefficients α and C_2 are derived from these equations as constant values for Blake–Kozeny and Burke–Plummer equations, respectively. These coefficients are described as:

$$\alpha = \frac{d_p^2}{150}\frac{\epsilon^3}{(1-\epsilon)^2}; \tag{6}$$

and

$$C_2 = \frac{3.5}{d_p}\frac{(1-\epsilon)}{\epsilon^3}. \tag{7}$$

In particle-resolved numerical simulations the source term from Eq. (4) depends on the pore size of the catalyst. As the pores have rather small diameter (10–100 nm), the momentum transported into the particle is negligible. Therefore, permeability is the main factor for the source term S_i and is written as:

$$S_i = \frac{\mu}{\alpha}v_i; \tag{8}$$

where α is permeability ($2.8 \cdot 10^{-8}$ m^2 for Al_2O_3 catalyst).

Generally, the flow inside the particle is laminar due to the resistance of the porous region. Therefore, the permeability coefficient α is the main factor that describes flow resistance in the porous media.

The energy equation for the inside part of the catalyst, where active chemical reactions are taking place is written in the following form:

$$\nabla \cdot \vec{v}\,(p + \rho h_m) = \nabla \cdot \left[-\left(\sum_j h_j\vec{J}_j\right) + \nabla T \cdot k_{\mathrm{eff}} + \bar{\bar{\tau}} \cdot \vec{v}\right] + S_h, \tag{9}$$

where h_m is mixture enthalpy; h_j and $\vec{J}_j$ are the specific sensible enthalpy and diffusion flux of species j, respectively; k_{eff} represents effective thermal conductivity of the isotropic porous media. S_h shows the heat from the external sources, which in this case is endothermic chemical reactions. The heat source was specified using the user-defined function (UDF).

The effective thermal conductivity of the porous material is shown as follows:

$$k_{\mathrm{eff}} = (1-\epsilon)k_m + \epsilon k_f \tag{10}$$

where k_m is the thermal conductivity of the catalyst solid material and k_f is the thermal conductivity of the mixture.

The mass transfer for each species is calculated using the diffusion equation for the turbulent flow:

$$\nabla \cdot (Y_j\, \rho\, \vec{v}) = -\nabla \cdot \vec{J}_j + R_j \tag{11}$$

where Y_j is the local mass fraction of element j; $\vec{J}_j$ represents the diffusion flux of species j and R_j is the overall production of species j by the chemical reactions.

The diffusion flux $\vec{J}_j$ due to the temperature and species concentration difference is calculated using Fick's law for each species j:

$$\vec{J}_i = -\left(\rho \cdot D_j^M + \rho \cdot D^t\right)\nabla Y_j - D_j^T \frac{\nabla T}{T} \tag{12}$$

where D_j^M is a mass diffusion coefficient for species j; D^t and D_j^T represent turbulent diffusivity and Soret thermal diffusivity for mixture species j, respectively. The mass diffusion coefficient outside the particle was treated according to the unity Lewis number condition. The mass diffusion inside the catalyst particle was calculated using the Bosanquet approximation. The detailed explanation of the calculation of mass diffusion coefficient could be found in the article [58]. The turbulent diffusion plays an insignificant role inside the catalyst particles due to the mostly laminar flow inside it. Therefore, only the first and third terms on the right-hand side of the equation define the diffusion.

The continuity, momentum, energy and species transport equations inside the porous catalyst zone are changed to include additional terms to account for porous media effect:

Continuity equation is written as:

$$\nabla \cdot (\epsilon \rho \vec{v}_{av}) = 0 \tag{13}$$

Momentum equation becomes:

$$\nabla \cdot (\rho \vec{v}_{av} \vec{v}_{av}/\epsilon) = -\nabla p + \nabla \cdot \bar{\bar{\tau}} + \rho \vec{g} - \left(\frac{\mu}{\alpha} + \frac{C_2}{2}\rho |\vec{v}_{av}|\right)\vec{v}_{av} \tag{14}$$

Energy equation is written as:

$$\nabla \cdot (\epsilon \vec{v}_{av}(\rho h_m + p)) = \nabla \cdot \left[-\left(\sum_j h_j \vec{J}_j\right) + k_{\text{eff}} \nabla T + \bar{\bar{\tau}} \cdot \vec{v}_{av}\right] + S_h \tag{15}$$

Species transport equation transform into:

$$\nabla \cdot (\rho \vec{v}_{av} Y_j) = -\nabla \cdot \vec{J}_j + \epsilon R_j \tag{16}$$

The $k-\omega$ SST turbulence model was chosen to describe the flow. Other studies report that it computes flow properties well with chemical reactions [59,60]. The transport equations are written as:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k v_i) = \frac{\partial}{\partial x_j}\left(\Gamma_k \frac{\partial k}{\partial x_j}\right) + G_k - Y_k; \tag{17}$$

$$\frac{\partial}{\partial t}(\rho \omega) + \frac{\partial}{\partial x_i}(\rho \omega v_i) = \frac{\partial}{\partial x_j}\left(\Gamma_\omega \frac{\partial \omega}{\partial x_j}\right) + G_\omega - Y_\omega + D_\omega, \tag{18}$$

where Γ_ω and Γ_k represent the effective diffusivity of ω and k, respectively; G_ω defines the generation of ω, G_k represents turbulent kinetic energy generation due to mean velocity gradients, Y_ω and Y_k represent dissipation of ω and k due to turbulence, respectively; D_ω represents cross-diffusion term.

Steam methane reforming is an endothermic process that includes three main reactions [61], which are presented below:

$$\mathrm{CH_4 + H_2O \Leftrightarrow CO + 3H_2} + 206.1\ \mathrm{kJ/mol}; \tag{19}$$

$$\mathrm{CO + H_2O \Leftrightarrow CO_2 + H_2} - 41.1\ \mathrm{kJ/mol}; \tag{20}$$

$$\mathrm{CH_4 + 2H_2O \Leftrightarrow CO_2 + 4H_2} + 165.0\ \mathrm{kJ/mol}. \tag{21}$$

For the present numerical simulation the model described by Hoang et al. was implemented [62]. That global reaction kinetics describes the rate of each chemical reaction written in Eqs. (19), (20) and (21), respectively, as follows:

$$r_1 = \frac{k_1\left(P_{\mathrm{CH_4}} \cdot P_{\mathrm{H_2O}} - \frac{P_{\mathrm{H_2}}^3 \cdot P_{\mathrm{CO}}}{K_{e1}}\right)}{P_{\mathrm{H_2}}^{2.5} \cdot Q_r^2}; \tag{22}$$

$$r_2 = \frac{k_2\left(P_{\mathrm{CO}} \cdot P_{\mathrm{H_2O}} - \frac{P_{\mathrm{H_2}} \cdot P_{\mathrm{CO_2}}}{K_{e2}}\right)}{P_{\mathrm{H_2}} \cdot Q_r^2}; \tag{23}$$

$$r_3 = \frac{k_3\left(P_{\mathrm{CH_4}} \cdot P_{\mathrm{H_2O}}^2 - \frac{P_{\mathrm{H_2}}^4 \cdot P_{\mathrm{CO_2}}}{K_{e3}}\right)}{P_{\mathrm{H_2}}^{3.5} \cdot Q_r^2}; \tag{24}$$

$$Q_r = 1 + K_{\mathrm{CO}} \cdot P_{\mathrm{CO}} + K_{\mathrm{H_2}} \cdot P_{\mathrm{H_2}} + K_{\mathrm{CH_4}} \cdot P_{\mathrm{CH_4}} + \frac{K_{\mathrm{H_2O}} \cdot P_{\mathrm{H_2O}}}{P_{\mathrm{H_2}}}, \tag{25}$$

where $P_{\mathrm{CH_4}}$, P_{CO}, $P_{\mathrm{H_2}}$, $P_{\mathrm{H_2O}}$ and $P_{\mathrm{CO_2}}$ are the mixture species partial pressure; k_1, k_2 and k_3 are the kinetic rate coefficients of reaction i, which equals $i = 1$ is Eq. (19), $i = 2$ is Eq. (20) and $i = 3$ is Eq. (21), respectively; K_{e1}, K_{e2} and K_{e3} represent the equilibrium constants of reaction i; $K_{\mathrm{CO}}, K_{\mathrm{H_2}}, K_{\mathrm{CH_4}}, K_{\mathrm{H_2O}}$ are the species adsorption constants.

Table 1
Arrhenius equation constants.

Pre-exponential factors			Activation energy		
k_{01}	k_{02}	k_{03}	ΔE_1	ΔE_2	ΔE_3
$9.048 \cdot 10^{11}$	$5.43 \cdot 10^5$	$2.14 \cdot 10^9$	209 500	70 200	211 500

Table 2
Van't Hoff equation constants.

Pre-exponential factors				Enthalpy of absorption			
K_{0CH_4}	K_{0H_2}	K_{0H_2O}	K_{0CO_2}	ΔH_{CH_4}	ΔH_{H_2}	ΔH_{CO}	ΔH_{CO_2}
$1.99 \cdot 10^{-3}$	$8.11 \cdot 10^{-5}$	$7.05 \cdot 10^{-9}$	$1.68 \cdot 10^4$	−36 650	−70 230	−82 550	85 770

The reaction kinetic rate coefficients (k_i) and species absorption constants (K_j) could be found using the Arrhenius and Van't Hoff equations, respectively, as shown below:

$$k_i = k_{0i} \cdot exp\left(\frac{-\Delta E_i}{RT}\right) \tag{26}$$

$$K_j = K_{oj} \cdot exp\left(\frac{-\Delta H_j}{RT}\right) \tag{27}$$

where k_{0i} and K_{oj} are the pre-exponential factors, ΔE_i is the thermal effect of the reaction and ΔH_j represents adsorption enthalpy of mixture species j.

The constants used for the Arrhenius and Van't Hoff equation are written in Tables 1 and 2, respectively.

The reaction kinetics described above is developed for the SMR reactions on catalyst particles with catalyst particle diameter (d = 1.75 mm). However, for the possible simulations with larger or smaller particles the reaction rate will change due to the introparticle diffusion limitation. To account for the intraparticle diffusion limitation, the efficiency coefficient η may be implemented in the form [46]:

$$R_{i,real} = \eta \cdot R_{i,intr} \tag{28}$$

2.3. Boundary conditions

The CFD simulations were performed using Ansys Fluent 2022 R2 software. The heat transfer characteristics of the steam methane reforming process were investigated for a wide range of operating conditions: inlet Reynolds number from 250 to 1500; mixture inlet temperatures from 700 K to 1100 K; catalyst particle diameter from 2.5 mm to 20 mm and various particle shapes. The ratio of $H_2O/CH_4 \approx 2$ is chosen as it provides high reaction rates and prevents carbon formation on catalyst surface [63,64]. Detailed boundary conditions, gas phase and catalyst properties are provided in Table 3. The adiabatic wall boundary condition was specified on the tube wall. Under these conditions, the sensible heat of the mixture, obtained from preheating, is converted into chemical energy via endothermic SMR reactions, without considering additional external heat supply.

The SIMPLE algorithm was used for pressure-velocity coupling. A pressure-based solver type was chosen. The momentum, continuity, species and energy equations were solved with the second discretization order to obtain the solution with higher accuracy. The species under-relaxation factor was set to 0.95 to ensure the solver stability. The convergence criteria were 10^{-6} for the continuity, species and energy. The solution was stopped after the convergence criteria were met or species mass fractions did not change for an extended time. Each solution was computed manually, but the automation of calculations using Python code may also be used [65].

The mixture density changed according to the incompressible ideal gas law, as the pressure difference was insignificant and temperature fluctuations were significant. The viscosity and thermal conductivity of the reaction mixture were calculated based on the mass-weighted mixing law. Specific heat dependence on temperature was written in a polynomial form with coefficients taken from Yuan et al. [66]. An industrial Ni-based catalyst on aluminum oxide support (Al_2O_3) was used for the present investigation. The main catalyst properties are: density of 1950 kg/m^3, effective thermal conductivity of 1 W/m K, specific heat 1000 J/kg K, void fraction of 0.45 [51].

3. Results

3.1. Model validation

The reaction kinetics proposed by Hoang et al. [62] was used in the current study to investigate the influence of steam methane reforming reactions on heat and mass transfer mechanisms. Experimental data from Hoang et al. [62] was used to validate the numerical model. This data is reliable for the current case as SMR reactions were investigated with the diffusion limitations for the particle size about 1.75 mm, which is notably higher than 0.2 mm particles without diffusion limitations [67].

Table 3
Summary of boundary conditions and physical properties.

Boundary conditions		
Parameter	Value/Range	Description
Inlet velocity	0.25–1.5 m/s	Superficial velocity range
Inlet temperature	700–1100 K	Pre-heated mixture range
Operating pressure	101 325 Pa (1 atm)	Constant atmospheric pressure
CH_4 mass fraction	0.3	Fixed composition
H_2O mass fraction	0.7	Fixed steam-to-methane ratio
Wall condition	Adiabatic	No heat transfer through walls
Outlet condition	Pressure outlet	Zero gauge pressure
Gas phase properties		
Density	Ideal gas law	Temperature/pressure dependent
Viscosity	Mass-weighted mixing law	Polynomial temperature dependence
Thermal conductivity	Mass-weighted mixing law	Polynomial temperature dependence
Specific heat	Polynomial function	Coefficients from [66]
Turbulence model	k-ω SST	Shear Stress Transport
Catalyst & Porous media properties		
Catalyst material	Ni/Al_2O_3	Industrial nickel-alumina
Density	1950 kg/m^3	Solid phase density
Specific heat	1000 J/(kg K)	Constant value
Effective thermal conductivity	1 W/(m K)	Isotropic property
Porosity (ϵ)	0.45	Void fraction
Permeability (α)	2.8×10^{-8} m^2	For Al_2O_3 support
Inertial resistance (C_2)	Eq. (7)	Calculated via Ergun equation
Particle shapes	Sphere, Cylinder, Raschig ring, 7-hole cylinder, Trilobe	Commercial catalyst geometries
Particle diameters	2.5–20 mm	Industrial size range

The reformer was a tube with d = 10 mm and a reaction zone length of 150 mm. The tube was filled with 8.98 g of spherical catalysts. The amount of Nickel-Alumina catalyst in the packed bed was around 39%. Therefore, the reaction rate was decreased not only by diffusion limitation but also by a low amount of catalyst. The mixture and reaction space were pre-heated to the initial temperature of $T = 973$ K and $T = 773$ K. Reformer operating pressure was kept at 150 kPa and H_2O/CH_4 ratio was 3. The reaction mixture residence time changed due to different mass flow rates on inlet.

The comparison of numerical simulation data with the experiment is presented in Fig. 4a. The dependence of methane conversion on the mixture residence time was analyzed. The average deviation is less than 5% with a maximum error resulting in 12% for $T = 973$ K and 4.2% for $T = 773$ K. The similarity between the numerical and experimental data is obvious. Moreover, the contours of CH_4 mass fraction and reformer temperature are shown in Fig. 4b. The scaling of the reformer in radial and axial directions was performed to improve the visual performance. Mixture temperature drops sharply at the beginning of the reaction zone due to the active heat consumption by SMR reactions, which is consistent with other studies [51]. The steep consumption of methane occurs throughout the packed bed with low concentrations near the reformer walls due to the higher reaction rate near the heated reactor wall.

In validation the packed bed was considered as a single porous zone, while in the numerical simulations the discrete particle approach was chosen. The implementation of a particle-resolved model instead the pseudo-homogeneous model is rather accurate when the heat source on the reformer wall is absent [68], which is the case for the presented investigation.

In addition, this CFD-model was validated for the real packed bed filled with spherical and cylindrical particles. The results of such validation are provided in authors' previous paper [68] where the same kinetic model was used for numerical simulation.

3.2. Heat transfer analysis

Numerical simulations were used to measure the influence of the steam methane reforming endothermic heat consumption on heat and mass transfer. The study aims to capture the dependence of reaction heat consumption on geometric and operating conditions. Thus, to provide clarity of the results, the only heat source for SMR reactions is the sensible heat of the mixture. That corresponds to the cases with pre-heated CH_4/H_2O mixture over a packed bed [39,69]. The dependence of reaction heat consumption on operating and geometric conditions is crucial for the catalyst choice and its performance.

The distribution of methane and hydrogen for inlet temperature $T_{in} = 1100$ K, velocity $\upsilon_{in} = 1.5$ m/s and particle diameter $d_p =$ 10 mm are presented in Fig. 5. The CH_4 mass fraction gradually decreases as the mixture flows from the inlet towards the outlet. The methane concentration is the lowest on the catalysts near the inlet and the highest on the last ones for all particle shapes. This observation is attributed to the higher temperature of the mixture near the inlet. A higher temperature facilitates the conversion greatly. The most active particle zones are near the catalyst surface, where the sharpest gradient is observed. This behavior is caused by the porous resistance that significantly decreases mixture velocity inside the catalyst particle [46]. Therefore, the reaction rate is at equilibrium in the middle of the catalyst. The H_2 concentrations are inverse to that of CH_4 as this is the main reaction product.

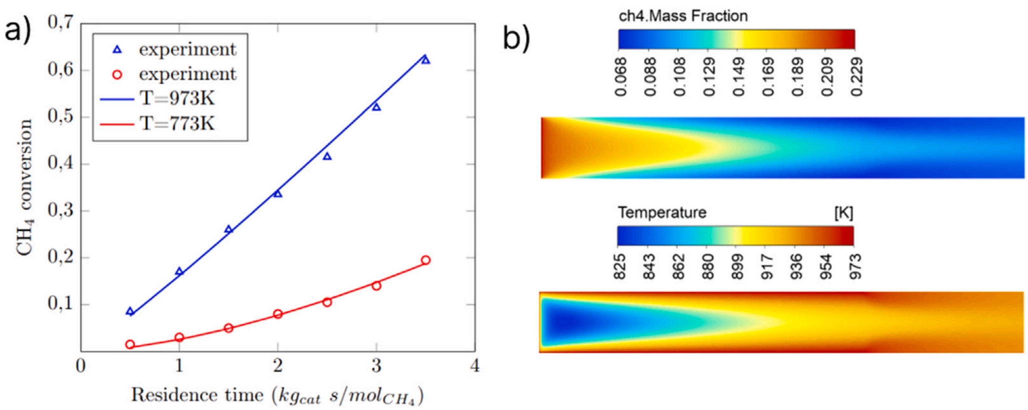

Fig. 4. (a) Methane conversion for various residence time and reactor temperatures, lines and hollow points represent numerical simulation and experimental data, respectively; (b) contours of methane mass fraction and temperature for $T = 973$ K and residence time of 3.5 kg_{cat}/mol_{CH_4}.

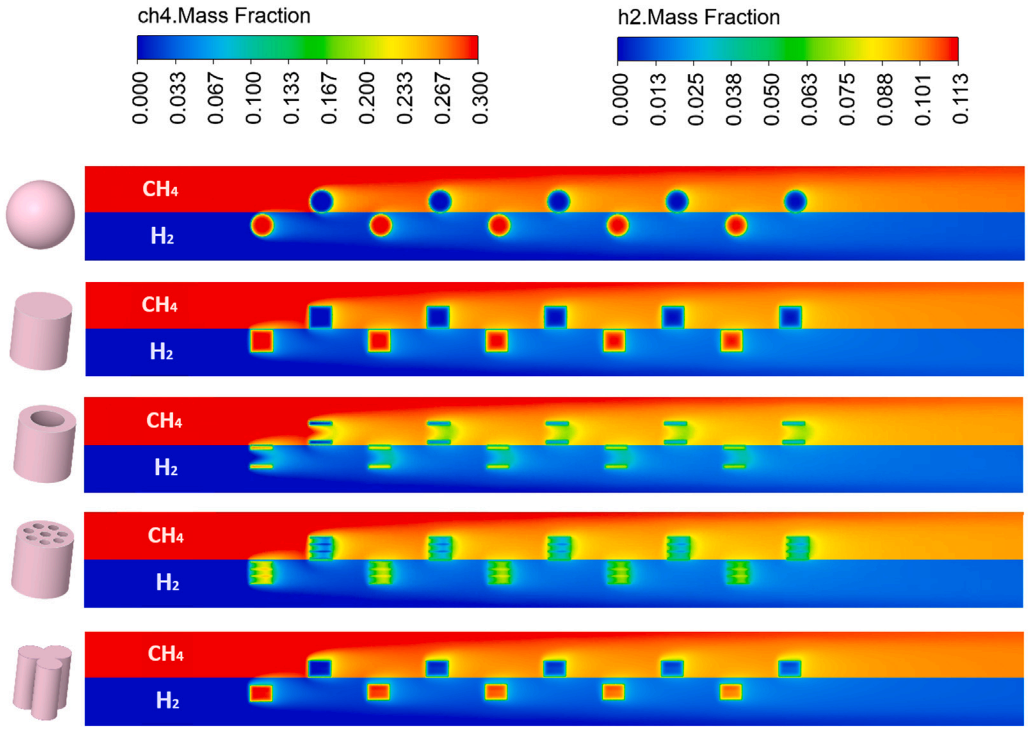

Fig. 5. Contours of methane and hydrogen distribution for various particles under: v_{in} = 1.5 m/s, T_{in} = 1100 K and d_p = 10 mm.

The temperature distributions for all particle shapes for inlet temperature T_{in} = 1100 K, velocity v_{in} = 1.5 m/s and particle diameter d_p = 10 mm are shown in Fig. 6. The temperature gradually decreases due to the endothermic SMR reactions as the mixture flows towards the outlet. The average temperature on the reactor outlet is higher for spherical and trilobe particles, which indicates lower average reaction rates. The catalyst volume greatly affects the size of zones with low temperatures. Moreover, the outlet average temperature is generally lower for catalysts with larger volumes. However, the surface area of the catalysts influences the heat transfer process to a much greater extent. Raschig ring and 7-hole cylinder particles have the same volume as spherical particles but very high surface area, which results in a temperature decrease comparable with cylindrical particles.

Methane conversion with respect to inlet temperature for inlet velocity of v_{in} = 1.5 m/s and particle diameter d_p = 20 mm is presented in Fig. 7a. The dependence is non-linear as the reaction rate increases rather rapidly with temperature [70]. The CH_4 conversion is higher for cylindrical-shaped particles and is lower for spherical and trilobe catalyst shapes. Total heat flow (W) is relatively high for cylindrical-shaped particles as they have a high surface area. As was shown earlier, a higher particle surface area enhances the heat transfer. The spherical shape is the most streamlined with a smaller surface area. However, the absorbed heat

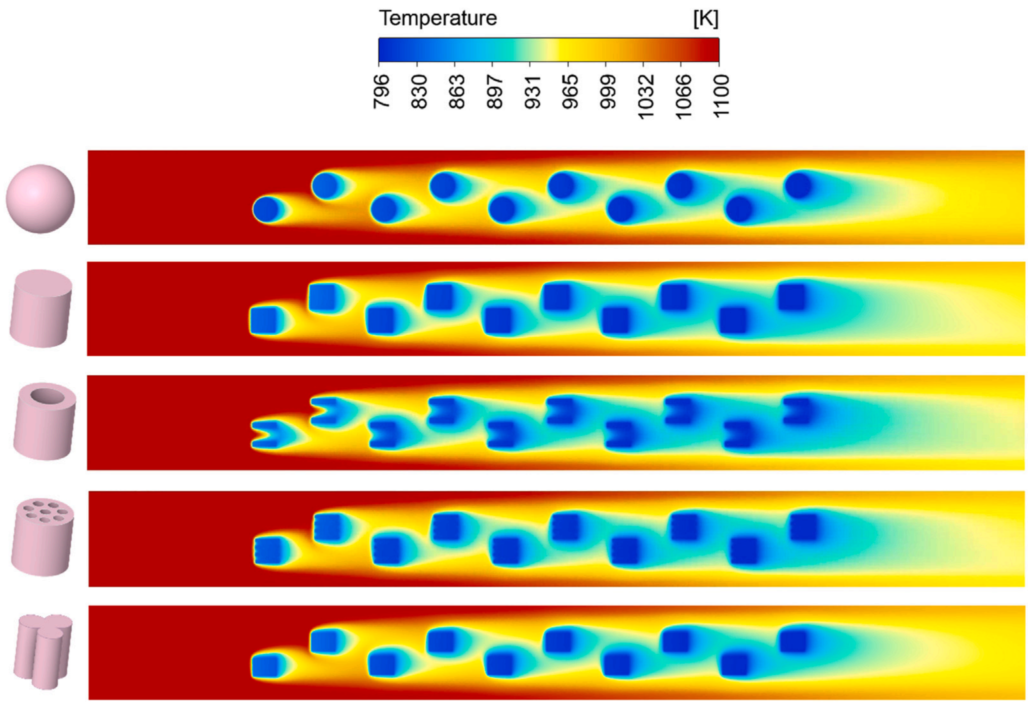

Fig. 6. Contours of temperature distribution for various particles under: υ_{in} = 1.5 m/s, T_{in} = 1100 K and d_p = 10 mm.

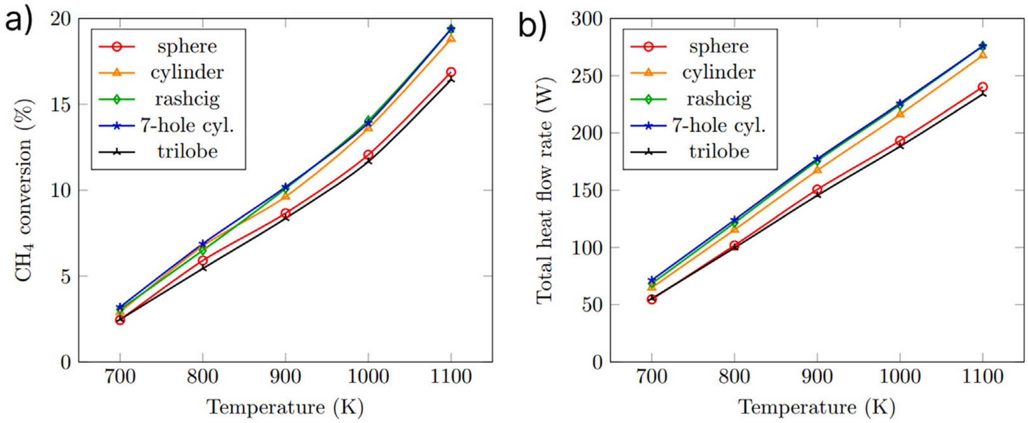

Fig. 7. Methane conversion and heat flow due to SMR reactions for various inlet temperatures, inlet velocity $\upsilon_{in} = 1.5$ m/s and particle diameter $d_p = 20$ mm.

on the trilobe is lower than on the sphere. Trilobe and sphere have practically the same volume but the external shape is rather different. The surface area of the trilobe is by 44% higher than that of the sphere and it is also externally shaped. Therefore, it is safe to conclude that externally shaped catalyst does not facilitate the conversion reaction. The relation of total heat flow due to steam methane reforming reactions to inlet temperature is shown in Fig. 7b. The dependence has a similar trend to CH_4 conversion on Fig. 7b, but the line shape is different due to the change in mass flow rate. The highest heat flow is 276 W for the Raschig ring and 7-hole cylinder and the lowest value is by 15% less for trilobe particles.

Methane conversion with respect to inlet velocity for inlet temperature of $T_{in} = 1100$ K and particle diameter $d_p = 20$ mm is shown in Fig. 8a. CH_4 conversion increases with lower inlet velocity. This behavior is characteristic and caused by the higher residence time of the mixture on catalysts. The general dependence is similar for all particle shapes. However, the increment in CH_4 conversion is higher under lower inlet velocity. The change of total endothermic effect of reaction related to inlet flow velocity is presented in Fig. 8b. The reaction efficiency is the highest for Raschig ring and 7-hole cylinders and is lowest for trilobe particle shape. The observation is consistent with Karthik and Buwa [36] who reported higher species conversion on internally shaped particles. The same particle efficiency distribution was observed for different inlet temperatures. Therefore, the shape of the particle influences the reaction efficiency in a similar manner both for various inlet temperatures and velocities.

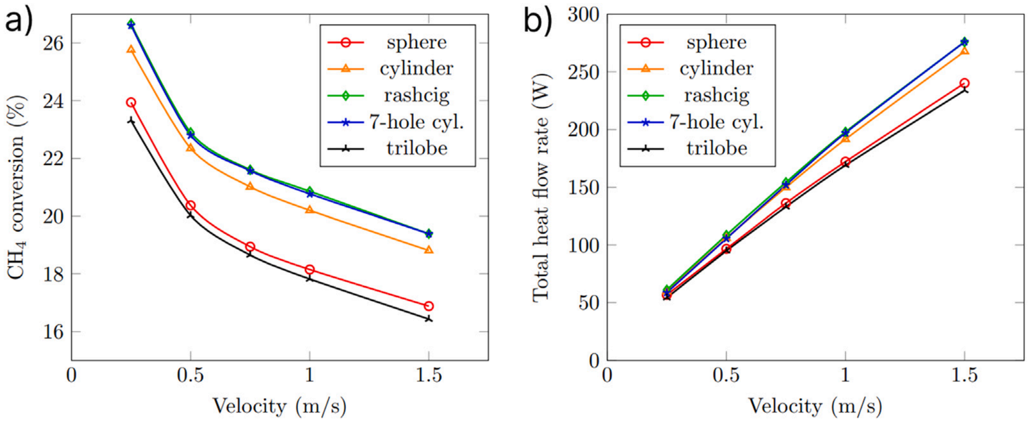

Fig. 8. Methane conversion and heat flow due to SMR reactions for various inlet velocities, inlet temperature T_{in} = 1100 K and particle diameter d_p = 20 mm.

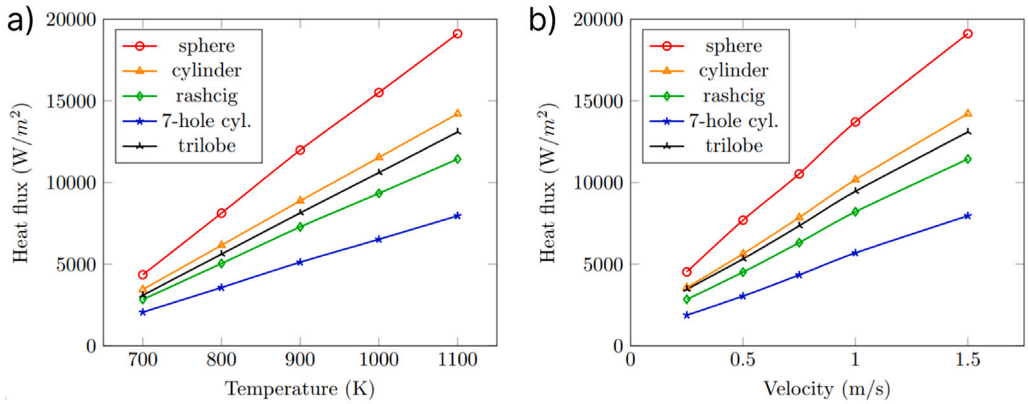

Fig. 9. Heat flux due to the reaction per m^2 of catalyst surface for particle diameter d_p = 20 mm under: (a) various inlet temperatures for inlet velocity v_{in} = 1.5 m/s and (b) various inlet velocities for inlet temperature T_{in} = 1100 K.

The total consumed heat flow (W) in the reactor may be used to compare the catalyst shapes. However, the catalysts investigated in the current study have the same characteristic size but different shapes and surface area. Therefore, using specific parameters is better to compare the effect of various catalyst shapes.

Fig. 9 shows the dependence of heat consumed by the reactions per m^2 of catalyst surface on inlet temperature and inlet velocity. The heat flux increases with higher temperature and the trend is almost linear. This effect is observed as an increase both in the reaction temperature leads to an increase in the sensible heat of the mixture and the reaction rate. The spherical catalysts have the lowest surface area and the most streamlined shape. Therefore, practically all of its surface area is easily accessed by the mixture, increasing the partial pressure of reactants inside the catalyst. In general, catalysts with higher surface area have lower heat flux. One exception is the trilobe particle, which has a slightly lower surface area than the cylindrical particle. That observation is explained by a higher volume of cylinder, which is by 1.5 times higher than that of the trilobe. Higher particle volume provides higher residence time that increases the CH_4 conversion. Therefore, the total volume of the particle also influences the heat flux.

The catalyst mass reflects the catalyst volume influence as the density is the same for all particles. Moreover, catalyst mass is the main parameter from the economic perspective as it determines packed bed cost. The mass heat flow per kg of catalyst for inlet velocity v_{in} = 1.5 m/s, d_p = 20 mm and various inlet temperatures is presented in Fig. 10. The Raschig ring performs best due to its high surface area and good flow characteristics. This catalyst shape allows the flow to pass both from the external surface and through the internal ring. The internal ring significantly reduces the diffusion limitation for reaction products from the internal catalyst volume. Lower diffusion limitation promotes a better species flow both from the inside of the catalyst and the main mixture flow. Thus, the internal ring reduces the catalyst mass and enhances the reaction rate at the same time. The second cost-effective shape is the 7-hole cylinder. Its mass and surface area are by 12.5% and 44%, respectively, higher than the Raschig ring, and it provides nearly the same amount of total heat flow (W). As shown in Fig. 5, the internal channels of a 7-hole cylinder contain zones with very low CH_4 concentration. These zones emerge due to the high flow resistance through narrow channels and, as a result, increase flow residence time but reduce the flow of fresh reactants from the mixture. As a result, the internal surface area of the 7-hole cylinder was utilized worse than that of the Raschig ring. Therefore, the catalysts with high surface area, low flow resistance

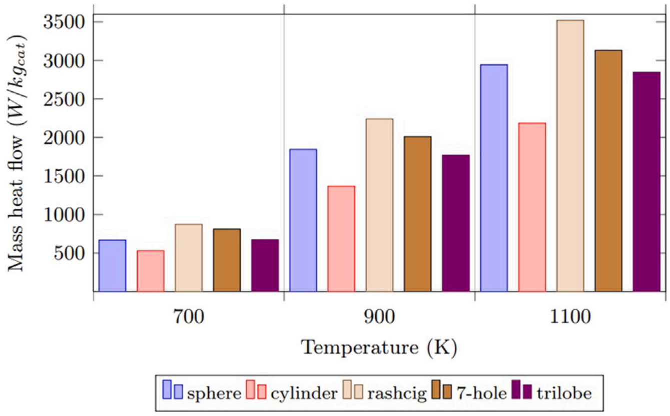

Fig. 10. Heat flow per kg of catalyst for various inlet temperatures, particle diameter $d_p = 20$ mm and inlet velocity $v_{in} = 1.5$ m/s.

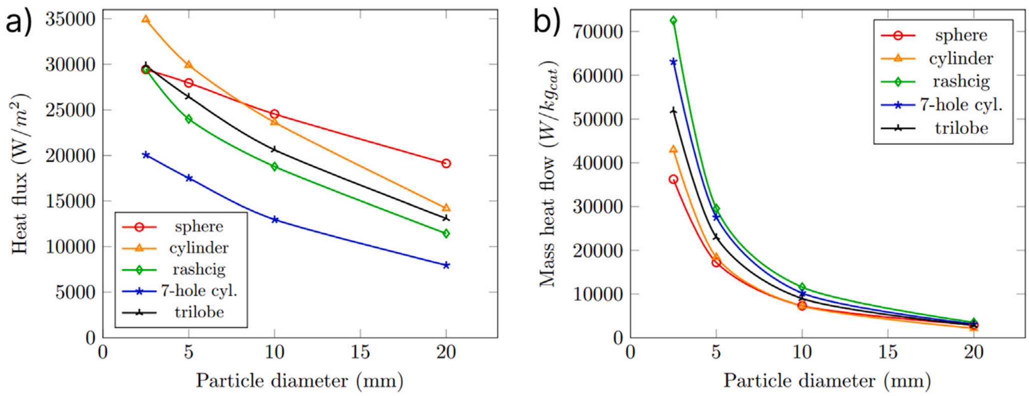

Fig. 11. Change of heat flow due to the SMR reactions from particle diameter per (a) m^2 of catalyst surface and (b) kg of catalyst under inlet velocity $v_{in} = 1.5$ m/s and inlet temperature $T_{in} = 1100$ K.

and streamlined shape have the best performance. The mass heat flow on the sphere is by only 6% less than on a 7-hole cylinder, but the sphere occupies a volume by 50% less. Therefore, packed beds with the same total heat flow (W) filled with spheres and 7-hole cylinders of large d_p will have a smaller volume for spheres. The packed bed with spherical particles could be used in cases where the reformer volume is of great importance.

The effect of particle diameter on the heat flow due to the reaction under inlet velocity of $v_{in} = 1.5$ m/s and inlet temperature of $T_{in} = 1100$ K is shown in Fig. 11. Four particle diameters d_p from 2.5 mm to 20 mm were considered as they are widely encountered in industry. With a decrease in particle diameter the mass heat flow and heat flux of the reaction increase. That trend is attributed to elevated temperatures inside the reactor as particles with lower diameters are able to convert less CH_4. As a result, the mixture has a higher average temperature and reaction rate. Heat flux per unit of catalyst surface (Fig. 11a) varies similarly for all particle shapes except for the spherical one. The behavior could be driven by constant shape streamlining, which leads to a linear function. The influence of particle diameter on mass heat flow per kg of catalyst (Fig. 11b) has a power function. From Fig. 11b it could be concluded that the Raschig ring and 7-hole cylinder have the best performance for all particle diameters.

The results for various catalyst shapes from operational parameters were presented earlier. To better visualize the results and make the difference between catalyst shapes more pronounced a shape factor ψ was introduced. The shape factor ψ shows the ratio of particle surface area, heat flux and mass heat flow to that of spherical particle. Three shape factors are written as follows:

$$\psi_F = F_{sphere}/F_{shape} \qquad \psi_q = q_{shape}/q_{sphere} \qquad \psi_Q = Q_{shape}/Q_{sphere} \tag{29}$$

where F_{sphere} and F_{shape} are the surface of spherical and another particle (cylinder, Raschig ring, 7-hole cylinder, trilobe), respectively; q_{shape} and q_{sphere} represents heat flux (W/m^2) on other particle and spherical one, respectively; Q_{shape} and Q_{sphere} are mass heat flow (W/kg) on the other particle and spherical one, respectively.

The dependence of total heat flow had a very weak dependence on inlet temperature and velocity as was presented in Figs. 7 and 8. Therefore, shape factor ψ is plotted in Fig. 12 as an average value for inlet temperatures and velocities for particle diameter $d_p = 20$ mm. From Fig. 12 it is clearly observed that maximum ψ_q was observed for spheres while maximum ψ_Q was for Raschig rings. The packed bed filled with spherical catalysts occupies a lesser volume with the same mass heat flow as Raschig rings. Therefore, it

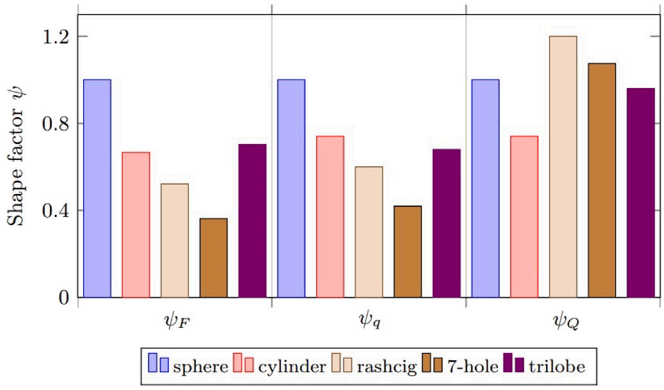

Fig. 12. Shape factor ψ for every particle for averaged inlet temperature T_{in}, inlet velocity v_{in} and particle diameter $d_p = 20$ mm.

may be used in cases, where the reformer volume is of great importance, for example, microreformer [71]. Whereas Raschig ring is a good shape from the economical perspective. The acquired findings are important for the choice of the catalyst shape for various industrial applications. The present study considers various shapes, but the direction of the particles is fixed. That is especially important for Raschig rings and 7-hole cylinders as their position in the flow may influence the heat transfer performance and flow characteristics considerably. Therefore, it is of great interest to study the catalyst shape performance for large-scale packed beds with random particle positions in the future.

4. Conclusion

This study presents the results of heat transfer analysis of the steam methane reforming process over Ni-based catalyst particles. The numerical simulations were developed to measure the effect of catalyst particle shape on the reaction heat consumption. Higher reaction heat consumption leads to higher H_2 yield, which indicates a better catalyst shape. Five catalyst shapes were considered: sphere, cylinder, Raschig ring, 7-hole cylinder and trilobe. The total heat flow rate (W) increased with higher inlet temperature and lower flow velocity. The total heat flow due to the reaction and CH_4 conversion were higher for Raschig rings and 7-hole cylinders, and the lowest was on trilobe particles. The heat flux (W/m^2) changed similarly from inlet velocity, temperature and particle diameter. Spheres with particle diameter d_p higher than 10 mm had the highest heat flux. The spherical particle has the smallest surface area and the most streamlined shape. Raschig ring and 7-hole cylinders had surface area by 92% and 176% higher than that of the sphere, and heat flux was lower by 67% and 140%, respectively. From an economic perspective, the highest heat flow of 3520 (W/kg) was observed on the Raschig rings and was by 12.5% lower for 7-hole cylinders. The Raschig ring and cylinders had practically the same total heat flow rate (W), but the mass of the 7-hole cylinder was higher by about 12.5%. The internal surface area of the 7-hole cylinder was utilized worse than the Raschig ring due to the high flow resistance and, consequently, lower mass flow of fresh reactants. The main parameters of catalyst shape that are beneficial for heat transfer are high surface area, low flow resistance and streamlined shape. Following these guidelines, one can select a better catalyst shape and create a new one with improved properties. The future study may include catalyst shape performance comparison for large-scale packed beds with random particle positions.

CRediT authorship contribution statement

Igor Karpilov: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Dmitry Pashchenko:** Writing – review & editing, Supervision, Software, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by Guangdong Provincial Key Laboratory of Materials and Technology for Energy Conversion (MATEC) under project KD2200047. There are no relevant financial or non-financial competing interests to report.

Data availability

No data was used for the research described in the article.

References

[1] Mohammad Mahbub Kabir, Sujit Kumar Roy, Faisal Alam, Sang Yong Nam, Kwang Seop Im, Leonard Tijing, Ho Kyong Shon, Machine learning-based prediction and optimization of green hydrogen production technologies from water industries for a circular economy, Desalination 567 (2023) 116992, http://dx.doi.org/10.1016/j.desal.2023.116992.

[2] Mona Wappler, Dilek Unguder, Xing Lu, Hendrik Ohlmeyer, Hannah Teschke, Wiebke Lueke, Building the green hydrogen market–current state and outlook on green hydrogen demand and electrolyzer manufacturing, Int. J. Hydrog. Energy 47 (79) (2022) 33551–33570, http://dx.doi.org/10.1016/j.ijhydene.2022.07.253.

[3] Dmitry Pashchenko, Integrated solar combined cycle system with steam methane reforming: Thermodynamic analysis, Int. J. Hydrog. Energy 48 (48) (2023) 18166–18176.

[4] D. Pashchenko, Performance evaluation of a combined power generation system integrated with thermochemical exhaust heat recuperation based on steam methane reforming, Int. J. Hydrog. Energy 48 (15) (2023) 5823–5835.

[5] Jeffrey Moore, Jon Durham, Andre Eijk, Enver Karakas, Rainer Kurz, Joseph Lesak, Marybeth McBain, Patrick McCalley, Leonid Moroz, Zahroof Mohamed, et al., Compressors and expanders, in: Machinery and Energy Systems for the Hydrogen Economy, 10.1016/B978-0-323-90394-3.00002-3, Elsevier, 2022, pp. 333–424.

[6] Jimmy Rojas, Shang Zhai, Eddie Sun, Vasudev Haribal, Sebastian Marin-Quiros, Amitava Sarkar, Raghubir Gupta, Matteo Cargnello, Will Chueh, Arun Majumdar, Technoeconomics and carbon footprint of hydrogen production, Int. J. Hydrog. Energy 49 (2024) 59–74, http://dx.doi.org/10.1016/j.ijhydene.2023.06.292.

[7] Nilesh Ade, Abdulrahman Alsuhaibani, Mahmoud M El-Halwagi, Henry Goyette, Benjamin Wilhite, Integrating safety and economics in designing a steam methane reforming process, Int. J. Hydrog. Energy 47 (9) (2022) 6404–6414, http://dx.doi.org/10.1016/j.ijhydene.2021.11.240.

[8] Ammar Ali Abd, Mohd Roslee Othman, Hasan Sh Majdi, Zuchra Helwani, Green route for biomethane and hydrogen production via integration of biogas upgrading using pressure swing adsorption and steam-methane reforming process, Renew. Energy 210 (2023) 64–78, http://dx.doi.org/10.1016/j.renene.2023.04.041.

[9] Marcello De Falco, Giulia Santoro, Mauro Capocelli, Giampaolo Caputo, Alberto Giaconia, Hydrogen production by solar steam methane reforming with molten salts as energy carriers: Experimental and modelling analysis, Int. J. Hydrog. Energy 46 (18) (2021) 10682–10696, http://dx.doi.org/10.1016/j.ijhydene.2020.12.172.

[10] Florian Pruvost, Schalk Cloete, Carlos Arnaiz del Pozo, Abdelghafour Zaabout, Blue, green, and turquoise pathways for minimizing hydrogen production costs from steam methane reforming with CO2 capture, Energy Convers. Manage. 274 (2022) 116458, http://dx.doi.org/10.1016/j.enconman.2022.116458.

[11] Nguyen Dat Vo, Jun-Ho Kang, Min Oh, Chang-Ha Lee, Dynamic model and performance of an integrated sorption-enhanced steam methane reforming process with separators for the simultaneous blue H2 production and CO2 capture, Chem. Eng. J. 423 (2021) 130044, http://dx.doi.org/10.1016/j.cej.2021.130044.

[12] Murtaza A Khan, Mohamed S Challiwala, Anuj V Prakash, Nimir O Elbashir, Conceptual modeling of a reactor bed of a nickel-copper bi-metallic catalyst for dry reforming of methane, Chem. Eng. Sci. 267 (2023) 118315, http://dx.doi.org/10.1016/j.ces.2022.118315.

[13] Puneet Kumar Chaudhary, Goutam Deo, Process and catalyst improvements for the dry reforming of methane, Chem. Eng. Sci. 276 (2023) 118767, http://dx.doi.org/10.1016/j.ces.2023.118767.

[14] Eugenio Meloni, Marco Martino, Vincenzo Palma, A short review on Ni based catalysts and related engineering issues for methane steam reforming, Catalysts 10 (3) (2020) 352, http://dx.doi.org/10.3390/catal10030352.

[15] Yixiong Lin, Chen Yang, Wei Zhang, Hiroshi Machida, Koyo Norinaga, Lattice Boltzmann study on the effect of hierarchical pore structure on fluid flow and coke formation characteristics in open-cell foam for dry reforming of methane, Chem. Eng. Sci. 268 (2023) 118380, http://dx.doi.org/10.1016/j.ces.2022.118380.

[16] Zhen-Yu Zhang, Ting Li, Zi-Yu Tang, Dan He, Jun-Jie Tian, Jia-You Chen, Tao Xie, Deep insight of the influence of pt loading content with catalytic activity on light-assisted dry reforming of methane, Chem. Eng. Sci. 274 (2023) 118710, http://dx.doi.org/10.1016/j.ces.2023.118710.

[17] Viacheslav Papkov, Boyan Zhang, Han Su, Haojie Chen, Dmitry Pashchenko, Numerical modeling of laminar flow over a porous cylinder under endothermic steam methane reforming reaction, International Journal of Heat and Fluid Flow 112 (2025) 109725, http://dx.doi.org/10.1016/j.ijheatfluidflow.2024.109725.

[18] Yi Ran Lu, Petr Nikrityuk, Scale-up studies on electrically driven steam methane reforming, Fuel 319 (2022) 123596, http://dx.doi.org/10.1016/j.fuel.2022.123596.

[19] Ali Akbar Darabadi Zare, Mortaza Yari, Hossein Nami, Farzad Mohammadkhani, Low-carbon hydrogen, power and heat production based on steam methane reforming and chemical looping combustion, Energy Convers. Manage. 279 (2023) 116752, http://dx.doi.org/10.1016/j.enconman.2023.116752.

[20] Ali Amini, Mohammad Hadi Sedaghat, Shaghayegh Jamshidi, Alireza Shariati, Mohammad Reza Rahimpour, A comprehensive CFD simulation of an industrial-scale side-fired steam methane reformer to enhance hydrogen production, Chem. Eng. Process.- Process. Intensif. 184 (2023) 109269, http://dx.doi.org/10.1016/j.cep.2023.109269.

[21] R. Tamilselvan, A. Immanuel Selwynraj, Model development for biogas generation, purification and hydrogen production via steam methane reforming, Int. J. Hydrog. Energy 50 (2024) 211–225, http://dx.doi.org/10.1016/j.ijhydene.2023.08.096.

[22] Wen-Jenn Sheu, Zei-Wei Hsu, Wei-Hsin Chen, Yen-Cho Chen, Investigation of steam methane reforming in a pd–ru membrane reactor with a counter-current configuration, Int. J. Hydrog. Energy 52 (2024) 938–952, http://dx.doi.org/10.1016/j.ijhydene.2023.04.009.

[23] Yi Ran Lu, Petr A. Nikrityuk, DEM-based model for steam methane reforming, Chem. Eng. Sci. 247 (2022) 116903, http://dx.doi.org/10.1016/j.ces.2021.116903.

[24] Mohsen Esfandiary, Seifolah Saedodin, Nader Karimi, On the effects of surface waviness upon catalytic steam reforming of methane in micro-structured reactors-a computational study, Int. J. Hydrog. Energy 52 (2024) 465–481, http://dx.doi.org/10.1016/j.ijhydene.2023.06.086.

[25] Jean-François Peloquin, Dany Francoeur, William Leclerc, Dino Mehanovic, Jean-François Dufault, Paul Camus, Ignacio Jorge Castellanos-Beltran, Nadi Braidy, Luc G Fréchette, Mathieu Picard, Electrified steam methane reforming microreactor, Int. J. Hydrog. Energy 49 (2024) 907–915, http://dx.doi.org/10.1016/j.ijhydene.2023.07.343.

[26] Yi Ran Lu, Petr A. Nikrityuk, Verification of a 0D model for the heat and mass transfer inside a moving spherical catalyst for steam methane reforming, Fuel 323 (2022) 124246, http://dx.doi.org/10.1016/j.fuel.2022.124246.

[27] Igor Karpilov, Dmitry Pashchenko, Effect of methane reforming before combustion on emission and calorimetric characteristics of its combustion process, Int. J. Hydrog. Energy 46 (76) (2021) 38106–38118, http://dx.doi.org/10.1016/j.ijhydene.2021.09.043.

[28] Philipp Wachter, Philipp Hödl, Juraj Raic, Christian Gaber, Martin Demuth, Christoph Hochenauer, Towards thermochemical recuperation applying combined steam reforming and partial oxidation of methane: Thermodynamic and experimental considerations, Energy Convers. Manage. 251 (2022) 114927, http://dx.doi.org/10.1016/j.enconman.2021.114927.

[29] Philipp Wachter, Christian Gaber, Martin Demuth, Christoph Hochenauer, Towards a recuperative, stationary operated thermochemical reformer: Experimental investigations on the methane conversion and waste heat recovery, Appl. Therm. Eng. 183 (2021) 116121.

[30] Arijit Ganguli, Viraj Bhatt, Hydrogen production using advanced reactors by steam methane reforming: a review, Front. Therm. Eng. 3 (2023) 1143987, http://dx.doi.org/10.3389/fther.2023.1143987.

[31] Jing Ma, Bo Jiang, Wang Si-ma, Kewei Yu, Zheng Lv, Yuming Gao, Dazhong Yuan, Dawei Tang, Deciphering high-efficiency solar-thermochemical energy conversion process of heat pipe reactor for steam methane reforming, Fuel 326 (2022) 124972, http://dx.doi.org/10.1016/j.fuel.2022.124972.

[32] A high temperature tubular reactor with hybrid concentrated solar and electric heat supply for steam methane reforming, Chem. Eng. J. 428 (2022) 132073, http://dx.doi.org/10.1016/j.cej.2021.132073.

[33] Jingyu Wang, Zongxin Liu, Changfa Ji, Lang Liu, Heat transfer and reaction characteristics of steam methane reforming in a novel composite packed bed microreactor for distributed hydrogen production, Energies 16 (11) (2023) 4347, http://dx.doi.org/10.3390/en16114347.

[34] V. Papkov, N. Shadymov, D. Pashchenko, Gas flow through a packed bed with low tube-to-particle diameter ratio: Effect of pellet roughness, Phys. Fluids 36 (2) (2024) 027127, http://dx.doi.org/10.1063/5.0183475.

[35] Zhihong Wu, Zhigang Guo, Jian Yang, Qiuwang Wang, Effect of diameter distribution of particles on methane steam reforming in multi-channel grille-sphere composite packed bed, Energy Convers. Manage. 265 (2022) 115764, http://dx.doi.org/10.1016/j.enconman.2022.115764.

[36] G.M. Karthik, Vivek V. Buwa, Effect of particle shape on fluid flow and heat transfer for methane steam reforming reactions in a packed bed, AIChE J. 63 (1) (2017) 366–377.

[37] Wenxin Xu, Zunhua Zhang, Yong Huang, Hang Qian, Gesheng Li, Effect of catalyst shapes on hydrogen production from exhaust gas-fuel reforming for NG engine: a mesoscale 3D simulation, Fuel 348 (2023) 128492.

[38] Yi Ran Lu, Petr A. Nikrityuk, Steam methane reforming driven by the Joule heating, Chem. Eng. Sci. 251 (2022) 117446, http://dx.doi.org/10.1016/j.ces.2022.117446.

[39] Majid Siavashi, Farzad Hosseini, Hamid Reza Talesh Bahrami, A new design with preheating and layered porous ceramic for hydrogen production through methane steam reforming process, Energy 231 (2021) 120952, http://dx.doi.org/10.1016/j.energy.2021.120952.

[40] Mohammad Rasoul Omidvar, Shoaib Khanmohammadi, Zahed Shababi, Ravinder Kumar, Performance assessment and exergy analysis of hydrogen production from natural gas in a petrochemical unit (a real case study), Int. J. Hydrog. Energy 52 (2024) 320–334, http://dx.doi.org/10.1016/j.ijhydene.2023.05.095.

[41] Lei Zheng, Matteo Ambrosetti, Francesca Zaio, Alessandra Beretta, Gianpiero Groppi, Enrico Tronconi, Direct electrification of Rh/Al2O3 washcoated SiSiC foams for methane steam reforming: An experimental and modelling study, Int. J. Hydrog. Energy 48 (39) (2023) 14681–14696, http://dx.doi.org/10.1016/j.ijhydene.2022.12.346.

[42] A.B. Shigarov, S.I. Uskov, D.I. Potemkin, P.V. Snytnikov, Experimental verification of kinetics and internal diffusion impact on low temperature steam reforming of a propane-methane mixture over Ni-based catalyst, Chem. Eng. J. 429 (2022) 132205, http://dx.doi.org/10.1016/j.cej.2021.132205.

[43] Jing Ma, Bo Jiang, Lin Li, Kewei Yu, Qian Zhang, Zheng Lv, Dawei Tang, A high temperature tubular reactor with hybrid concentrated solar and electric heat supply for steam methane reforming, Chem. Eng. J. 428 (2022) 132073, http://dx.doi.org/10.1016/j.cej.2021.132073.

[44] Dmitry Pashchenko, Anton Eremin, Heat flow inside a catalyst particle for steam methane reforming: CFD-modeling and analytical solution, Int. J. Heat Mass Transfer 165 (2021) 120617, http://dx.doi.org/10.1016/j.ijheatmasstransfer.2020.120617.

[45] Mostafa Pourali, Javad Abolfazli Esfahani, Mohammad Amin Sadeghi, Kyung Chun Kim, Jeff Gostick, Simulation of methane steam reforming in a catalytic micro-reactor using a combined analytical approach and response surface methodology, Int. J. Hydrog. Energy 46 (44) (2021) 22763–22776, http://dx.doi.org/10.1016/j.ijhydene.2021.04.124.

[46] Dmitry Pashchenko, Intra-particle diffusion limitation for steam methane reforming over a Ni-based catalyst, Fuel 353 (2023) 129205, http://dx.doi.org/10.1016/j.fuel.2023.129205.

[47] Małgorzata Wójcik, Łukasz Szabłowski, Olaf Dybiński, Comparison of mathematical models of steam methane reforming process for the needs of fuel cells, Int. J. Hydrog. Energy 52 (2024) 965–982, http://dx.doi.org/10.1016/j.ijhydene.2023.08.293.

[48] Martin Kutscherauer, Scott D Anderson, Sebastian Böcklein, Gerhard Mestl, Thomas Turek, Gregor D Wehinger, A conjugated heat and mass transfer model to implement reaction in particle-resolved CFD simulations of catalytic fixed bed reactors, Eng. Appl. Comput. Fluid Mech. 18 (1) (2024) 2292100, http://dx.doi.org/10.1080/19942060.2023.2292100.

[49] Steffen Flaischlen, Thomas Turek, Gregor D. Wehinger, Local structure effects on hydrodynamics in slender fixed bed reactors: Spheres and rings, Chem. Eng. J. 475 (2023) 146342, http://dx.doi.org/10.1016/j.cej.2023.146342.

[50] Junjie Chen, Yehao Yu, Numerical modelling for design of microchannel reactors: Application to hydrogen production from methanol by steam reforming, Int. J. Hydrog. Energy 50 (2024) 1526–1540, http://dx.doi.org/10.1016/j.ijhydene.2023.11.061.

[51] Zhihong Wu, Zhigang Guo, Jian Yang, Qiuwang Wang, Numerical investigation of hydrogen production via methane steam reforming in a novel packed bed reactor integrated with diverging tube, Energy Convers. Manage. 289 (2023) 117185, http://dx.doi.org/10.1016/j.enconman.2023.117185.

[52] Igor Karpilov, Dmitry Pashchenko, Steam methane reforming over a preheated packed bed: Heat and mass transfer in a transient process, Therm. Sci. Eng. Prog. 42 (2023) 101868, http://dx.doi.org/10.1016/j.tsep.2023.101868.

[53] Dmitry Pashchenko, Experimental study of methane reforming with products of complete methane combustion in a reformer filled with a nickel-based catalyst, Energy Convers. Manage. 183 (2019) 159–166, http://dx.doi.org/10.1016/j.enconman.2018.12.102.

[54] Yi Ran Lu, Dmitry Pashchenko, Petr A. Nikrityuk, A new semiempirical model for the heat and mass transfer inside a spherical catalyst in a stream of hot CH4/H2O gases, Chem. Eng. Sci. 238 (2021) 116565, http://dx.doi.org/10.1016/j.ces.2021.116565.

[55] Wei Wang, Yong Cao, Tsubasa Okaze, Comparison of hexahedral, tetrahedral and polyhedral cells for reproducing the wind field around an isolated building by LES, Build. Environ. 195 (2021) 107717, http://dx.doi.org/10.1016/j.buildenv.2021.107717.

[56] Amira Neni, Yacine Benguerba, Marco Balsamo, Alessandro Erto, Barbara Ernst, Djafer Benachour, Numerical study of sorption-enhanced methane steam reforming over Ni/Al2O3 catalyst in a fixed-bed reactor, Int. J. Heat Mass Transfer 165 (2021) 120635, http://dx.doi.org/10.1016/j.ijheatmasstransfer.2020.120635.

[57] Lei Cai, Tianzhi He, Yanlei Xiang, Yanwen Guan, Study on the reaction pathways of steam methane reforming for H2 production, Energy 207 (2020) 118296, http://dx.doi.org/10.1016/j.energy.2020.118296.

[58] Igor Karpilov, Viacheslav Papkov, Dmitry Pashchenko, Comparative analysis of diffusion mechanisms inside porous media for steam methane reforming over Ni-Al2O3 catalyst, Int. Commun. Heat Mass Transfer 159 (2024) 108322.

[59] Hongbum Choi, Shin Hyuk Kim, Joongmyeon Bae, Sai PR Katikaneni, Aqil Jamal, Aadesh Harale, Stephen N Paglieri, Jay H Lee, CFD analysis and scale up of a baffled membrane reactor for hydrogen production by steam methane reforming, Comput. Chem. Eng. 165 (2022) 107912, http://dx.doi.org/10.1016/j.compchemeng.2022.107912.

[60] Anthony G. Dixon, Local transport and reaction rates in a fixed bed reactor tube: Endothermic steam methane reforming, Chem. Eng. Sci. 168 (2017) 156–177, http://dx.doi.org/10.1016/j.ces.2017.04.039.

[61] Nico Jurtz, Matthias Kraume, Gregor D. Wehinger, Advances in fixed-bed reactor modeling using particle-resolved computational fluid dynamics (CFD), Rev. Chem. Eng. 35 (2) (2019) 139–190, http://dx.doi.org/10.1515/revce-2017-0059.

[62] D.L. Hoang, S.H. Chan, O.L. Ding, Kinetic and modelling study of methane steam reforming over sulfide nickel catalyst on a gamma alumina support, Chem. Eng. J. 112 (1) (2005) 1–11, http://dx.doi.org/10.1016/j.cej.2005.06.004.

[63] Haotian Zhang, Zhuxing Sun, Yun Hang Hu, Steam reforming of methane: Current states of catalyst design and process upgrading, Renew. Sustain. Energy Rev. 149 (2021) 111330, http://dx.doi.org/10.1016/j.rser.2021.111330.

[64] Dmitry Pashchenko, Ivan Makarov, Carbon deposition in steam methane reforming over a Ni-based catalyst: Experimental and thermodynamic analysis, Energy 222 (2021) 119993, http://dx.doi.org/10.1016/j.energy.2021.119993.

[65] Vyacheslav Papkov, Nikita Shadymov, Dmitry Pashchenko, CFD-modeling of fluid flow in ansys fluent using python-based code for automation of repeating calculations, Internat. J. Modern Phys. C 34 (09) (2023) 2350114, http://dx.doi.org/10.1142/S0129183123501140.

[66] Qinyuan Yuan, Rong Gu, Jing Ding, Jianfeng Lu, Heat transfer and energy storage performance of steam methane reforming in a tubular reactor, Appl. Therm. Eng. 125 (2017) 633–643, http://dx.doi.org/10.1016/j.applthermaleng.2017.06.044.

[67] Jianguo Xu, Gilbert F. Froment, Methane steam reforming, methanation and water-gas shift: I. intrinsic kinetics, AIChE J. 35 (1) (1989) 88–96, http://dx.doi.org/10.1002/aic.690350109.

[68] Viacheslav Papkov, Dmitry Pashchenko, CFD-modeling of steam methane reforming via particle-resolved and pseudo-homogeneous models: comparative analysis with experimental verification, Fuel 393 (2025) 134906.

[69] Nenad Zecevic, Nenad Bolf, Advanced operation of the steam methane reformer by using gain-scheduled model predictive control, Ind. Eng. Chem. Res. 59 (8) (2020) 3458–3474, http://dx.doi.org/10.1021/acs.iecr.9b06260.

[70] Samrand Saeidi, András Sápi, Asif Hussain Khoja, Sara Najari, Mariam Ayesha, Zoltán Kónya, Bernard Baffour Asare-Bediako, Adam Tatarczuk, Volker Hessel, Frerich J Keil, et al., Evolution paths from gray to turquoise hydrogen via catalytic steam methane reforming: Current challenges and future developments, Renew. Sustain. Energy Rev. 183 (2023) 113392, http://dx.doi.org/10.1016/j.rser.2023.113392.

[71] Sherzod Abdullaev, Nusratilla Rajabovich Barakayev, Barno Sayfutdinovna Abdullaeva, Umid Turdialiyev, A novel model of a hydrogen production in micro reactor: conversion reaction of methane with water vapor and catalytic, Int. J. Thermofluids 20 (2023) 100510, http://dx.doi.org/10.1016/j.ijft.2023.100510.